PATENT APPLICATION · PROVISIONAL DRAWINGS

Cross-Botanical Terpene Matching & Multisensory Cannabis Experience Curation System

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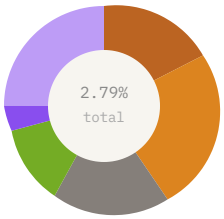
PATENT PENDING

> **FIG. 3** — Worked example: *Mike's Bomba* (Glentucky Family Farm, Sonoma County, California). Illustrates the complete system pipeline from SC Labs GC-MS terpene lab analysis (INPUT) through the cross-botanical matching engine to two parallel outputs: (A) a strain-matched essential oil blend formula specifying constituent essential oils and drop counts in a 2ml standard batch with diffusion protocol, and (B) Spotify-compatible audio parameter values for curated playlist generation. Demonstrates application of the terpene source-independence principle and the 15–30 minute pre-loading temporal sequence.

01 INPUT — SC LABS GC-MS TERPENE ANALYSIS



Mike's Bomba

Glentucky Family Farm · Sonoma County, CA · SC Labs COA

β-Caryophyllene	1.24%
d-Limonene	0.83%
α-Humulene	0.41%
Myrcene	0.22%
Linalool	0.09%
TOTAL TERPENES (DRY WEIGHT)	2.79%

CROSS-BOTANICAL MATCHING ENGINE
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02 OUTPUT A — STRAIN-MATCHED ESSENTIAL OIL BLEND (2ML / 40 DROPS)

Black Pepper *(Piper nigrum)* **7 drops**

→ β-Caryophyllene (dominant), α-Humulene (secondary) · 47% of formula

Lemon *(Citrus limon)* **5 drops**

→ d-Limonene (primary constituent ~65–70%) · 33% of formula

Sage *(Salvia officinalis)* **2 drops**

→ α-Humulene, camphor, 1,8-cineole · 13% of formula

Lavender *(Lavandula angustifolia)* **1 drop**

→ Linalool (~35–45%), β-Ocimene (secondary) · 7% of formula

Total **15 drops · 0.75ml active in 2ml batch (FCO/argan carrier)**

PRE-LOAD PROTOCOL — TEMPORAL SEQUENCING INSTRUCTION

Add blend to ultrasonic cold-water diffuser. Diffuse for **15–30 minutes** prior to cannabinoid consumption. Do not administer simultaneously. Pre-loading interval enables A2a receptor priming prior to THC engagement of A2a-CB1 heteromeric receptor complexes.

AUDIO PARAMETER TRANSLATION MODULE

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03 OUTPUT B — SPOTIFY AUDIO PARAMETER PRESCRIPTION

TEMPO	KEY / MODE	ENERGY
72–82 BPM	**D♭ Minor**	**0.31**
β-Caryophyllene CB2 dominance → lower range grounding	CB2 + GABA-A engagement → Aeolian / Dorian	Grounding profile with citrus lift offset from d-Limonene
VALENCE	ACOUSTICNESS	DANCEABILITY
0.44	**0.62**	**0.48**
Dark base (β-Caryophyllene) tempered by d-Limonene 5-HT1A warmth	Organic warm instrumentation preference	Moderate groove; not driving

INSTRUMENTATION RECOMMENDATION

Deep bass · cello · muted electric guitar · sparse low-end percussion · minimal sustained pads

MECHANISM — RECEPTOR PATHWAY ANNOTATION

β-Caryophyllene (1.24%, dominant): Selective CB2 agonist and TRPA1 modulator. Anti-inflammatory signalling via NF-κB pathway suppression. Primes CB2 receptor populations during the 15–30 min pre-load interval. Reflected in: low BPM (65–80), minor key, low energy/valence parameters.

d-Limonene (0.83%, secondary): 5-HT1A partial agonist; dopaminergic modulation. Anxiolytic offset that moderates the dominant CB2 grounding profile. Source-independent: d-Limonene from lemon EO is structurally and pharmacologically identical to d-Limonene from cannabis. Reflected in: moderate valence uplift (0.44 vs. predicted 0.30 from β-Caryophyllene alone).

α-Humulene (0.41%, tertiary): CB2/PPARγ co-agonist; reinforces anti-inflammatory action of β-Caryophyllene. Source: hops, coriander, sage EO. Reflected in: sustained minor key, restrained energy.

Pre-loading temporal mechanism: Inhalation of the EO blend activates olfactory-limbic, trigeminal-autonomic, and pulmonary-systemic delivery pathways simultaneously. A2a receptor priming during the interval modifies A2a-CB1 heteromeric complex conformation, producing altered allosteric signalling upon subsequent THC-CB1 engagement compared to unprimed (naive) receptor populations. This temporal distinction is the core patentable novelty.